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Virtual Blood Bank Mobile Application to Enhance Emergency Response for  
Maternal Care in Ethiopia

## Software Requirements Specification

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## Document Approval

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## Definitions, Acronyms, and Abbreviations

This subsection provides the definitions of all terms, acronyms, and abbreviations required to properly interpret this SRS. This information is intended to ensure clarity and a common understanding among all stakeholders.

### Definitions:

#### Term Definition

**Actor:** An entity, either a human user or an external system, that interacts with the Virtual Blood Bank application.

**Blood Supply Chain:** The end-to-end process of managing blood from donation and collection to storage, distribution, and transfusion.

**District:** A defined administrative or geographical area in Ethiopia containing a cluster of healthcare facilities that will collaborate using the application.

**Healthcare Facility:** A hospital, clinic, or health center that provides medical services, stores blood, and is a participant in the Virtual Blood Bank network.

**Healthcare Professional:** The primary user of the application, including doctors, nurses, hospital administrators, and blood bank technicians.

**mHealth:** The use of mobile and wireless technologies to support the achievement of health objectives.

**Postpartum Hemorrhage (PPH):** Severe bleeding after childbirth, a leading cause of maternal mortality.

**Use Case:** A written description of a sequence of actions that a user performs with the system to achieve a specific goal.

**User-Centered Design (UCD):** A design philosophy and process in which the needs, wants, and limitations of end users of a product are given extensive attention at each stage of the design process.

**Virtual Blood Bank:** The core concept of this project: a software system that acts as a digital information hub for real-time blood inventory and requests, without holding any physical blood units itself.

### Acronyms and Abbreviations

#### Acronym: Definition

**API:** Application Programming Interface

**AWS:** Amazon Web Services

**BSIS:** Blood Safety Information System

**CCB:** Change Control Board

**CR:** Change Request

**DB:** Database

**EBTBS:** Ethiopian Blood and Tissue Bank Service

**EULA:** End-User License Agreement

**GCP:** Google Cloud Platform

**GUI:** Graphical User Interface

**HIS:** Health Information System

**HTTPS:** Hypertext Transfer Protocol Secure

**JSON:** JavaScript Object Notation

**MMR:** Maternal Mortality Rate

**MTBF:** Mean Time Between Failures

NFR: Non-Functional Requirement  
NGO: Non-Governmental Organization  
OS : Operating System  
PPH: Postpartum Hemorrhage  
QA: Quality Assurance  
RAM: Random-Access Memory  
RBAC: Role-Based Access Control  
SDD: Software Design Document  
SMS: Short Message Service  
SRS: Software Requirements Specification  
UAT: User Acceptance Testing  
UI: User Interface  
UML: Unified Modeling Language  
UX: User Experience  
VBB: Virtual Blood Bank  
WHO: World Health Organization

#### Requirement Identifier Convention

All specific requirements, use cases, and other detailed items within this document are assigned a unique identifier for traceability. This identifier follows the format: VBB-TYPE-MODULE-####.

VBB: Stands for Virtual Blood Bank, the project identifier.

TYPE: A code indicating the type of item. Examples include:

FUN: Functional Requirement  
NFR: Non-Functional Requirement  
UI: User Interface Requirement  
HW: Hardware Interface Requirement  
COMM: Communications Interface Requirement  
DB: Database Requirement  
INV: Inverse Requirement  
DC: Design Constraint  
UC: Use Case

MODULE: A code indicating the functional area or module the item belongs to. Examples include:

UM: User Management  
BIM: Blood Inventory Management  
BRQ: Blood Requesting  
PERF: Performance  
SEC: Security

####: A three-digit sequential number to ensure the identifier is unique.

## DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included. We have adequately cited and referenced the

original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

**Group NO Team Date:** xx-xx-2021

## 1. Introduction

This Software Requirements Specification (SRS) document outlines all the requirements of the virtual Blood Bank mobile application. The main aim of this software is to help solve the critically high maternal mortality rate in rural Ethiopia, especially the deaths which could be caused by postpartum hemorrhage (PPH) caused by the inefficient and inaccessible blood supply chains. Currently, maternal mortality rates in Ethiopia are 401 deaths per 100,000 live births and the national acceptable annual blood requirement is 200,000-400,000 units that is not fulfilled annually because of a fragmented paper-based system and real-time communication between blood banks and rural health facilities.

The Virtual Blood Bank app will be used as a localized, district-by-district information center, which should allow monitoring blood supply and demand in real-time among already established hospitals and clinics. It will also enable the healthcare professionals (doctors, nurses, and hospital administrators) to seek specific types of blood, order it in their surrounding cooperating facilities, monitor the status of the requests, and get essential notifications about the quantity of stock. Through its ability to provide inter-hospital cooperation and the efficient management of resources, the system would assist in ensuring the availability of safe blood to obstetric crises within the required time, thus, making a direct impact on a statistically significant decrease in maternal death cases.

The document is the basic agreement that exists between the stakeholders and the development team, which contains all the information needed by the software engineers to design, build, and test the presented software product. It will outline the functional and non-functional requirements, technical limitations, and the overall system architecture, and make sure that a secure, reliable, and contextually well-adapting cross-platform mobile application is developed.

### 1.1 Purpose

This Software Requirements Specification (SRS) aims to clearly specify all functional and non-functional requirements of the mobile application of Virtual Blood Bank. The document should be the blueprint of development team according to which the design, implementation, and verification of the project should be planned. It specifies the software functionality, functionality, and the limitations according to which it is going to take place.

❖ Specifically, this SRS will:

- ✓ Create a shared knowledge between all stakeholders in the project about the functionalities and shortcomings of the software.
- ✓ Specify the software engineers in detail so that it forms the main source of designing and establishing the application.
- ✓ Provide a foundation of testing and quality assurance processes as well as to confirm that the developed software remains within all the requirements given.
- ✓ Serve as a control document of managing the changes on the requirements during the project lifecycle.



- ✓ Assist in future maintenance and improvement by well documenting the initial purpose and functionality of the system.
- ❖ This SRS is targeted at the following audience:
  - ✓ Software Engineers/Developers: To be acquainted with what exactly is required to be constructed to guide architectural design, code and integration.
  - ✓ UI/UX Designers: UI/UX Designers To inform the user interface and user experience design according to the functional and non-functional requirements.
  - ✓ Quality Assurance / Testing Team: To design all-inclusive test plans, test cases, and perform testing in accordance with requirements.
  - ✓ Project Managers: To oversee project scope, schedule, resources and compare progress against specified deliverable s.
  - ✓ Stakeholders (e.g., Ethiopian Government Health Officials, Ethiopian Blood Bank Representatives): To review and accept the functionality of the agreed system, to make sure that it fits their vision and requirements.
  - ✓ Domain Experts (e.g., Healthcare Professionals in rural clinics): To test whether the requirements translate into the operation needs and address the identified problem.
  - ✓ Technical Writers: To prepare user manuals and other support documentation to the application.

## 1.2 Scope

The following SRS outlines the specification of one and specific software product: the Virtual Blood Bank Mobile Application. Virtual Blood Bank mobile application is created as a distributed and real-time information system to improve the efficiency of blood supply chain in certain rural districts in Ethiopia. The application will offer an online service that will allow licensed medical workers (doctors, nurses, and hospital administrators) to:

- Real-time Blood Inventory Visibility: Show the current inventory of the various major blood groups (A+, A-, B+, B-, AB+, AB-, O+, O-,) within their own facility as well as other cooperating and recognized healthcare facilities within their respective district. This visibility will be updated within 5 minutes after any change in stock (e.g. transfusion, new donation entry).
- Inter-Facility Blood Requesting: Allow users to place a request electronically seeking a particular type and amount of blood in a different cooperating facility, and the source of the request and destination clearly audited.
- Request Status Tracking: Provide the requesting and supplying facilities the ability to track the status of a blood request in real-time (e.g. pending, accepted, in transit, received, rejected, etc.)
- Automated Notifications and Alerts: Issue context-specific in-app notifications (and possibly SMS with facilities with weak internet access) on events like: new blood requests, acceptance/ rejection of requests, dispatch of blood units and critically low stock levels (customizable thresholds based on blood type/facility).
- Secure User and Facility Profile Management: Supply the establishment, verification, and administration of user accounts connected with particular healthcare institutions, which guarantees secure role-dependent access.

- Mapping of Donation Centers: Add mapping feature, which will take the potential voluntary blood donor to the closest active blood collection center with up-to-date working hours and contact.

**The Virtual Blood Bank mobile application will not:**

- ◇ Has any physical blood storage, processing, or transportation infrastructure. Any physical blood storage and transportation will continue to be dependent on the available capacities of the hospitals and local transport systems (e.g., cooler bags, which will be transported by the motorbike/ambulance with fuel, as in the operational costs).
- ◇ Make any medical tests, screening, or make any blood units. By using these essential activities, the current recognized entities such as the Ethiopian Blood and Tissue Bank Service will remain in performing such activities.
- ◇ Participate in the actual piece of blood collection with the donors. Its purpose is purely educative to direct the donors to the current collection sites.
- ◇ Incorporate courses in full medical training of healthcare practitioners on transfusion procedures or overall medical procedures. The training of the application will be restricted to the basic user onboarding to the functionalities of the application.
- ◇ Eliminate maternal mortality in Ethiopia. Its intervention is particularly aimed at the minimization of maternal mortality due to insufficient and timely blood transfusion, which, in the first place, is the result of postpartum bleeding (PPH).
- ◇ Be made or optimized to support non-maternal emergency, though incidental utility of some of the information can occur. Motherly health crisis is the primary user stories and features.
- ◇ Act as a complete national maternal healthcare system or complement all the available healthcare information systems in Ethiopia. It is a district-level intervention program.
- ◇ Offer face-to-face interfaces to pregnant mothers or their families. Healthcare professionals are the major end-users.
- ❖ The Virtual Blood Bank mobile application would be applied to fill the existing logistical gaps in blood provision to rural maternal care in Ethiopia.

**The objectives, goals, and benefits, in question, are stated as follows:**

- Reducing Preventable Maternal Deaths due to PPH: A quantifiable decrease in the rates of maternal mortality in target rural districts due to the inadequate supply of sufficient and safe blood in 30 minutes of request initiation to blood dispatch to address urgent obstetric emergencies.
- Enhance Resource Productiveness in Blood: This can be achieved by streamlining the blood resource use by minimizing waste through expiry and maximizing the chances of not having shortages in one facility and excess in another. This is to achieve an approximate of 20-30% decrease in blood products wastage in participating facilities.
- Enhance Inter-Hospital Blood Sharing and Cooperation: This approach ensures a smooth, digital-based blood sharing system among collaborating hospitals and clinics in a district, and avoids the use of informal, time-intensive systems such as family replacement donors in case of emergencies. This is aimed at ensuring that 90 percent of blood transfer requests are met in the district.
- Empower Healthcare Provider Decision-Making: equip healthcare providers with real-time, actionable data (blood availability, request status, alerts) to allow them to make critical decisions

regarding patient care faster and more confidently during maternal emergencies and to reduce decision-making time associated with the access of blood by 50 percent.

- Expand Voluntary Blood Donation Accessibility: Showing the donation centers on the map, contribute indirectly to the increased number of the voluntary donor base and the more regularity of the donation process, which will allow the potential donor to donate blood when they have the right.

### 1.3 Overview

This SRS is organized into three main sections.

- ✓ **Section 1 (Introduction)**: Provides an overview of the SRS, including its purpose, scope, and organization.
- ✓ **Section 2 (General Description)**: Describes the overall context of the product, including its relationship to other systems, its key functions, user characteristics, and general constraints. This section provides background information to better understand the specific requirements.
- ✓ **Section 3 (Specific Requirements)**: Contains the detailed functional and non-functional requirements for the software. This section will cover all functionalities, performance criteria, security standards, and other essential system attributes.

## 2. General Description

This section provides a high-level overview of the Virtual Blood Bank mobile application, describing the general factors that influence the product and its requirements. The aim here is to establish a broader context, making the detailed functional and non-functional requirements specified in subsequent sections easier to understand, rather than listing specific requirements themselves.

### 2.1 Product Perspective

The Virtual Blood Bank mobile application is developed to be a single mHealth intervention. It should be used to supplement, not to substitute the current healthcare infrastructure and other related initiatives in Ethiopia. Particularly, it will coexist and act autonomously with the wider Ethiopian national blood management system with which is running the Blood Safety Information System Enhancement (BSIS) project. Although the BSIS is concerned with the national inventory and management at the macro level through the Ethiopian Blood and Tissue Bank Service, the Virtual Blood Bank app will offer a specific, granular, and localized solution, as it will be focused on inter-hospital collaboration on a district-by-district level in rural areas.

The application is a virtual overlay over the current physical blood storage units within the rural hospitals and clinics. It majorly handles data on these current physical inventories of blood. It does not add new blood storage and transportation physical infrastructure. Even though this system will be independent in its main purpose, its architecture will be able to take into account the future possible points of integration with larger national or regional health information systems, as long as opportunities will be provided. The app will be on regular mobile phones and will utilize available internet or cellular network connection, which is common in the regions of the target.

### 2.2 Product Functions

On the top level, the Virtual Blood Bank mobile application will be fulfilling some essential functions which will mainly be focused on the exchange of information and the effective coordination of the available resources concerning the blood products. These functions include:

- **Real-Time Blood Availability Display:** It is a display that gives a clear and real-time picture of the available type of blood in various cooperating healthcare facilities within a particular district.
- **Streamlined Blood Request Management:** Facilitating the process of healthcare providers making fast requests to other facilities to obtain required types of blood, and the lifecycle of these requests (e.g., acceptance, rejection, dispatch).
- **Automated Alerting and Notification System:** Providing a timely alert about urgent blood requests, a status update on a request, and a warning about a low inventory of blood that is critically low.
- **Donor Information & Navigation:** Assists potential voluntary blood donors by giving them information about where to find the closest active blood collection centers as well as instructions.
- **Greater Data Openness to Local Supply:** Adding to the more transparent and practical knowledge of blood supply and demand trends on district level.

## 2.3 User Characteristics

The main direct users of the Virtual Blood Bank mobile app will be trained medical workers working in the rural Ethiopian clinics and hospitals. This user group includes:

- **Doctors:** They shall use the application to confirm the stock of blood to meet patient demands especially those with complications such as postpartum bleed, and request blood.
- **Nurses:** Nurses will be the users of the app since they are often engaged in patient care, strategic management of the ward-level resources, and coordination with facility blood storage so that they could check the inventory in real-time and request its fulfillment.
- **Hospital Administrators / Blood Bank Technicians (at the facility level):** It is the duty of these people to take care of proper records of blood stocks, control of inventory and reception and dispatching of blood units using the application.

These users are expected to have a basic amount of technical literacy, as well as an overall acquaintance with smartphone applications. The user interface (UI) and user experience (UX) design of the application will thus focus on the ultimate level of ease of use, clear navigation as well as professional and clean look. The goal of this approach is to ensure that the learning curve is minimum and an in-depth technical training is not required, so that healthcare personnel would quickly adopt the technology and begin using it efficiently, even when this personnel lacks prior knowledge of the sophisticated digital tools. Although mothers and their families are the end users of the application, they do not necessarily use this particular application.

## 2.4 General Constraints

This sub section will describe the broad considerations and general constraints that will determine the design and development decisions on the Virtual Blood Bank mobile application. These are not specific, detailed requirements but high-level considerations which the developers should keep in mind to make the system viable and effective in the target environment.

**The general constraints of the developer will be constrained by, but not restricted to, the following:**

1. **Regulatory Policies:** The design and data processing mechanisms of the application have to be based on the strict compliance with Ethiopian national regulations and policies related to blood management, medical data privacy, and digital health programs. This involves adherence to any directive of the Ministry of Health or of the Ethiopian Blood and Tissue Bank Service.
2. **Hardware Limitations:** This application should be implemented to work on various current Android smartphones (since they are most common) and possibly enterprise-distributed iOS gadgets. It means that it has to be considered in relation to different screen sizes, processing capabilities, and battery duration, especially in the settings where the presence of the charging infrastructure may be erratic.
3. **Interfaces to other Applications:** Although the system is a stand-alone application at first, the system design is expected to look into the future where the system may be integrated with other healthcare information systems (HIS) that might be implemented or even made mandatory in Ethiopian hospitals. Such interoperability in the future should not be excluded in design decisions.
4. **Reliability Requirements:** Since the health emergencies of mothers are very critical, high level of reliability of the application is required. The data about the availability of blood and request features should always be accurate and be available even in the tough network conditions. The system must also be able to withstand the disruption of connectivity.
5. **Application Importance:** This is a life critical application given the direct effect it has on the mortality of the mother. The failure of the system, the lack of accuracy of data, or slowness of information flow may have dire results. This increases the requirement on the strong error handling, data integrity and fast performance.
6. **Safety and Security Issues:**
  - ✓ **Data Security:** This is of top priority to store sensitive patient information as well as data of blood inventory. The system should have high-quality encryption, access control tools, and secured communication protocols as a measure of avoiding unauthorized access, data breaches as well as manipulation.
  - ✓ **Application Safety:** The architecture should avoid the occurrence of undesirable functionality or bugs that may result in wrong medical diagnosis or diversion of important resources.
7. **Resource-Limited Space:** Rural Ethiopian conditions are characterized by:
  - ✓ **Unstable Network Connectivity:** The application needs to be bandwidth friendly and with an offline data capturing feature and followed up by synchronization when network connectivity is regained.
  - ✓ **Low Power Supply:** architecture must be made with regard to efficiency of mobile phone battery.
  - ✓ **Technical Infrastructure:** The application needs to be based on a wide local server infrastructure at each clinic; a cloud based backend is an inevitable constraint.
8. **Cost-Effectiveness:** Development, deployment, and operational costs need to be kept to the minimum in order to achieve a sustainable development of the solution and its appropriate scalability in the context of limited resources in national budget and possible NGO funding framework.
9. **User Technical Acumen (as a design constraint):** Since, as mentioned in Section 2.3, healthcare professionals may have different levels of technical literacy as a baseline, the user interface and workflows must be extraordinarily intuitive and user-friendly to reduce the number of detailed settings or jargon usage.

## 2.5 Assumptions and Dependencies

The subsection recognizes such significant factors, which though are not direct design constraints on the software itself, may substantially influence the validity of the requirements that are expressed in this SRS in case they alter. These are outside forces which the project presumes will remain true in regards to successful development and implementation of the Virtual Blood Bank mobile application. In case any of these assumptions turn out to be false or any dependence changes are made, the requirements contained in this SRS would require re-consideration and possibly adjustment.

❖ The main assumptions and dependencies are:

1. Availability of Compatible Mobile Devices: It is also assumed that the main direct consumers (healthcare providers in rural clinics and hospitals) will be able to have consistent access to compatible smartphone devices that can support the application. To a large extent, this hinges on the domination of Android devices in the target areas, and there may be the possibility of considering iOS distributed to enterprises by specific facilities. Unavailability of devices or transition to unsupported operating systems would lead to a re-consideration of the mobile platform strategy.
2. Basic and Intermittent Internet/Cellular Access: The app will require at least intermittent access to basic internet or cellular network connectivity in the intended rural districts. Although optimized to meet the low-bandwidth and negligible offline capabilities, scalable data synchronization and real-time communication rely upon this underlining network availability. The total or critical loss of connectivity would essentially change the functionality of the application in the form of a virtual blood bank.
3. Active Collaboration and Implementation by Healthcare Facilities: The willingness of the rural clinics and hospitals involved in the inter-hospital collaboration and real-time information exchange to implement, use, and continuously maintain their blood inventory data in the application is the key to the success of the entire process. The absence of engagement or inputting of the data would make the concept of the virtual blood bank meaningless.
4. Operation of Existing Physical Blood Storage Infrastructure: The project supposes that there will exist, be maintained and be operational the current physical blood storage units in the participating hospitals and clinics. The information on these physical resources is handled by the application; it cannot substitute these resources. The degradation or the end of these physical facilities would compromise the usefulness of the application.
5. Sustainability Funding of Operational Costs: The existing operational costs (e.g., cloud hosting, possible need of SMS gateway services and basic maintenance) are assumed to be funded by the government, NGOs or international health organizations, as shown in the economic feasibility study. The shortcomings in the form of continued funding would affect the long term sustainability and accessibility of the implemented system.
6. Availability of Established Blood Collection Services: The donor mapping capability of the application relies on the fact that there are existing and licensed blood collection centers and the banks capable of collecting physical blood. Any major shift in their functioning or scope of operation would impinge on the applicability of this aspect.



7. Ethiopian Blood and Tissue Bank Service (EBTBS) and Ministry of Health support: It is expected that EBTBS and the Ministry of Health will assist in this localized mHealth project, be willing to provide any regulatory assistance needed (where applicable to share donor center data), and endorse the mHealth plan in order to make it more adoption-friendly by healthcare facilities.
8. Local Technical Support Availability: Although the system is set to be robust in nature, in the long term the success of the system regarding its operation is pegged on the eventual availability of local technical support or expertise to help solve minor problems or offer support to the users.

### 3. Specific Requirements

This part specifies the functional and non-functional specifications of the Virtual Blood Bank mobile application. These requirements are defined to a degree that would allow software designers to design the system and quality assurance personnel to design extensive test plans. All requirements outline externally observable capacities or attributes of the system such as the inputs, outputs and functions of the system. The requirements can be distinctly recognized, validated, and in line with the overall objectives and goals of the project as established in the earlier sections.

#### 3.1 User Management and Authentication

This sub sub section outlines the specifications to be followed when managing user accounts and accessing the Virtual Blood Bank mobile application. The system will facilitate different user roles among the healthcare professionals.

##### 3.1.1 User Registration and Account Creation

- VBB-FUN-UM-001: The system will enable an assigned administrative user (i.e., district administrator or lead of a particular healthcare facility) to add new user accounts to healthcare professionals.
  - ✓ Input: Full Name (string up to 100 characters), E-mail (string, unique, valid format), Phone Number (string, unique, valid Ethiopian format), Facility ID (pre-existing system ID, numeric), Role (enum: Doctor, Nurse, Administrator, Blood Bank Technician), Temporary Password (system generated string).
  - ✓ Output: Account created; temporary password generated by this system sent to the email or phone number registered by the user.
- VBB-FUN-UM-002: New users shall be forced to change their temporary passwords upon the initial successful access to the system.
  - ✓ Input: Temporary Password (string), New Password (string, at least 8 characters, including uppercase/lowercase, numbers, special characters), Confirm New Password (string, it must be the same as New Password).
  - ✓ Output: The confirmation of password change; the access to application functionalities.
- VBB-FUN-UM-003: The system will uniquely identify a healthcare facility to which any user account is affiliated.
  - ✓ Output: Profile of user showing related facility.

### 3.1.2 User Logging in and Sessions.

- VBB-FUN-UM-004: The system will mandate the users to identify themselves with either their registered email address or phone number and their password.
  - ✓ Input: Password (string), Email/Phone Number (string).
  - ✓ Output: User was able to log in and access the application or receive an invalid credential error.
- VBB-FUN-UM-005: The system will be configured to have 30 minutes inactivity during which the user will be automatically logged out.
  - ✓ Response: User redirected to log-in screen.
- VBB-FUN-UM-006: The system will have a manual functionality that enables users to log out of their session.
  - ✓ Input: The user clicking the option of Logout.
  - ✓ Output: User sent to login page and session ended.
- VBB-FUN-UM-007: The system must have the option of a Remember Me to continue with the login credentials of 7 days on a particular device, unless they are manually log out.
  - ✓ Input: When a user logs in, he/she chooses the checkbox Remember Me.
  - ✓ Output: User was automatically logged into the application when it started and during the 7-day time span.

### 3.1.3 Password Management

- VBB-FUN-UM-008: The system will enable authorized users to alter their password.
  - ✓ Input: Current Password (string), New Password (string, at least 8 characters, with uppercase and lowercase, number, special character), Confirm New Password (string, needs to be the same as New Password).
  - ✓ Output: Password change has been made, or there has been an error of an incorrect current password or an unmatched new passwords.
- VBB-FUN-UM-009: The system will offer a system of forgetting password where the user can reset his/her password through a verified email or via phone number.
  - ✓ Input: Registered Email/Phone number (string).
  - ✓ Output: Password reset link/code is sent to a verified contact; when the user is verified a new password is entered.
- VBB-FUN-UM-010: The system will at rest encrypt all user passwords.
  - ✓ Output: N/A (internal security measure).

### 3.1.4 User Profile Management

- VBB-FUN-UM-011: The system will show the full name of the user, the facility where the user represents and his/her position on the profile page.
  - ✓ Output: User profile details.
- VBB-FUN-UM-012: The system must enable the users to change their complete name and phone number in their profile.
  - ✓ Input: New Full Name (string, up to 100 characters), New Phone Number (string, in the Ethiopian format).
  - ✓ Output: Profile update confirmation.

## 3.1 External Interface Requirements

The subsection provides the description of the requirements related to the interfaces between the Virtual Blood Bank mobile app and the users and any necessary hardware interfaces.



### 3.1.1 User Interfaces

This sub-section explains the logic nature of the interface between the Virtual Blood Bank mobile app and its immediate users who are the healthcare professionals in rural Ethiopian institutions. The system will offer graphical user interface (UI) that is to be used on touchscreen mobile devices.

- VBB-UI-001: The system will include a concise, intuitive, and consistent graphical user interface (UI) on all screens and functions.
  - ✓ Testable: The consistency will be measured in the form of following a set of specified design (color palette, typography, iconography) defined in the Software Design Document (SDD).
- ✓ VBB-UI-002: The system will employ a professional aesthetic that is clean and has enough contrast to be readable in different light conditions (e.g., bright sunlight, low clinic lighting, etc.).
  - ✓ Testable: Contrasting ratios of text and background elements shall comply with WCAG 2.1 AA requirements of legible text.
- VBB-UI-003: The system will use common patterns of mobile navigation (e.g. bottom navigation bar with primary functions, hamburger menu with secondary), that will have the least learning curves.
  - ✓ Testable: Adherence to a set of mobile UI/UX standards (e.g., Material Design, Apple Human Interface Guidelines).
- VBB-UI-004: The system will be written using clear and concise language with the utilization of jargon where feasible. The text will be mainly in Amharic and/or English and will have the functionality of language selection.
  - ✓ Testable: It will be verified by native Amharic speakers that all text displayed to the users is clear and exactly as intended.
- VBB-UI-005: The system will visually acknowledge (e.g. loading spinners, success/error messages) every interaction with the system and system business processes.
  - ✓ Testable: Within 2 seconds, there will be a visual response in terms of feedback to every action taken by a user.
- VBB-UI-006: The system will have portrait and landscape orientation to support the best experience of the system in different mobile devices.
  - ✓ Testable: It will be possible that application layout and functionality will be maintained and can be utilized in both orientations.
- VBB-UI-007: The system will make the touch targets of interactive elements (buttons, links, input field) large (at least 48x48 dp) to suit the needs of full range of finger size and possible fine motor skill issues.
  - ✓ Verifiable: The size of the elements of the UI will be measured according to design requirements.
- VBB-UI-008: The system will be designed to minimize the amount of steps necessary to perform essential actions which may involve searching blood or requesting it, and the target actions should be not more than three steps.
  - ✓ Verifiable: UAT will involve the user task flow analysis, which will ensure conformity.
- VBB-UI-009: The system must have some basic accessibility features, including adjustable text sizes and high-contrast displays, to accommodate users with minor visual impairments.
  - ✓ Testable: The accessibility settings of the device will be tested to ensure that basic functionality is achieved.

### 3.1.2 Hardware Interfaces

The Virtual Blood Bank mobile application is created as a purely software based solution to operate on the regular, commercially available mobile computers. The system does not have any direct control or interfacing requirements of specialized hardware components, other than the basic functionality of a smartphone.

Specifically:

- VBB-HW-001: The system will be able to use the graphical output of the device through the screen.
- VBB-HW-002: The system will use the touchscreen of the device as a user input.
- VBB-HW-003: The system will be configured to use the internal memory (RAM) and the internal storage of the device to execute the applications and hold temporary data.
- VBB-HW-004: The system will use the cellular modem and/or Wi-Fi radio on the device to communicate with its backend services via the network.
- VBB-HW-005: The system will use the GPS or location features of the machine to identify the geographical position of the user to be able to use the system to find features like locating donation centers within the neighbourhood.
- VBB-HW-006: The system will use the notification features of the devices (e.g., vibration, sound, on-screen notifications) to provide vital user-related notifications.

The client does not communicate with a external medical device, barcode scanners or any other proprietary hardware. Its interplay with the hardware is restricted to consuming services offered by underlying mobile operating system and the standard components of the device it is running on.

### 3.1.3 Software Interfaces

Any mandatory interactions with other software products or external application systems required by the scope of the project or upper requirements of the project are specified in this subsection. These are interfaces that the Virtual Blood bank mobile application will have to use or rather integrate with and not design options of internal components.

#### 3.1.3.1 Mobile Operating Systems Services.

The mobile application will be Virtual Blood Bank and it will be integrated with the services offered by underlying mobile operating systems in order to be effective. These are not third party applications but basic software layers that are needed to execute mobile applications.

#### **Name: Android Operating System Services.**

- Mnemonic: ANDROID-OS
- Specification Number: N/A (see official Android Developer documentation on a variety of APIs)
- Version Number: Android 8.0 (Oreo) and further.
- Source: Google (AOSP)
- Purpose: The app will make use of Android OS services to support the main functions, including:
  - ✓ Push Notifications (e.g., Firebase Cloud Messaging in case of emergency notifications).
  - ✓ Location Services (GPS to locate blood charity centers).
  - ✓ Local Storage (to store data hard that allows it to operate offline).
  - ✓ Network Access (Wi-Fi, Cellular Data to communicate with the backends).
  - ✓ Hardware (e.g. vibration).
- Interface Definition: The interface will be based on the standard Android SDK APIs (e.g., LocationManager, NotificationManager, ConnectivityManager, SQLite to local DB). The content and format of the message will be according to specifications of Android API.

**Name: iOS Operating System services.**

- Mnemonic: IOS-OS
- Specification Number: N/A (see official Apple Developer documentation on a number of APIs)
- Version Number: iOS 13.0 or more (assuming enterprise distribution is sought).
- Source: Apple Inc.
- Purpose Like Android, the application will utilize iOS services in:
  - ✓ Push (e.g., Apple Push Notification service).
  - ✓ Location Services (Core Location framework).
  - ✓ Local Storage (data storing).
  - ✓ Network Access.
  - ✓ Device Hardware Access.
- Interface Definition: All will be married via standard iOS SDK APIs (e.g. Core Location, UserNotifications, Core Data/Realm to local database). The content and format of messages will be in compliance with appropriate iOS API.

**3.1.3.2 Custom Backend API**

The mobile app will be connected to a specially designed back-end server to handle the data of the blood inventory, users, requests and notifications. This is the backend of the entire Virtual Blood Bank and is not physically connected to the mobile client.

Name: Virtual Blood Bank Backend API

- Mnemonic: VBB-BACKEND-API
- Specification Number: Will be described in a specific Backend API Specification Document.
- Version Number: 1.0 (First Issue)
- Source: Was developed in the context of this project.
- Purpose: This API offers data and business logic services required in the mobile app. It will handle:
  - ✓ Authentication and authorization of the users.
  - ✓ Blood inventory data storage and retrieval.
  - ✓ Blood request (creation, update, tracking).
  - ✓ Provision of real time notifications.
  - ✓ Healthcare facility data management.
  - ✓ Data management of blood donation center.
- Interface Definition: The interface will be characterized as a RESTful API that will communicate via HTTPS and use JSON (JavaScript Object Notation) as the content of the messages. Certain endpoints, request formats (GET, POST, PUT, DELETE), the request/response body format, and authentication (e.g., OAuth2, JWT) will be explicitly detailed in the Backend API Specification Document.

**3.1.3.3 Potential Future Interface: Ethiopian Blood and Tissue Bank Service (EBTBS) BSIS**

Though not an essential part of the initial scope, the project considers the presence and current implementation of the Ethiopian Blood Safety Information System Enhancement (BSIS) project, which is the Ethiopian Blood and Tissue Bank Service.

**Name: Ethiopian Blood Safety Information System (BSIS)**

- Mnemonic: EBTBS-BSIS
- Specification Number: It is not known at the moment, to be supplied by EBTBS.
- Version Number: Unknown at the present time.
- Source: Ethiopian Blood and Tissue Bank Service in association with HABTech Solutions.

- Purpose: Future integration may entail the exchange of macro-level information of blood supply or blood donor information, after official API and collaboration accord is reached. Virtual Blood Bank app is an inter-hospital collaborative tool but is not obliged at present to be directly connected with BSIS to perform its main functionality.
- Interface Definition: Not required at this point, since this is a possible future dependency and is not a requirement at this point. Any integration in the future would demand formal agreements and specifications of API of EBTBS.

### 3.1.4 Communications Interfaces

This sub section identifies communication protocols that the Virtual Blood Bank mobile application will directly interface directly, particularly in case they are custom or have specifics that need to be implemented by the designers. In the case of standard protocols that are used transparently (ex: by web services, such as HTTP/HTTPS), no reference to the standard is necessary unless the custom interaction is needed.

- VBB-COMM-001: The system will mainly employ HTTPS (Hypertext Transfer Protocol Secure) when carrying out all communications between mobile application client and the VBB-BACKEND-API (see 3.1.3.2).
  - ✓ Purpose To provide secure, encrypted and authenticated communication over possibly unsecured networks (Wi-Fi, cellular data) to preserve sensitive medically and user data during transit.
  - ✓ Details of Protocol: Encryption: TLS 1.2 or above. Normal HTTP methods (GET, POST, PUT, DELETE) will be utilized.
- VBB-COMM-002: The system is expected to provide SMS (Short Message Service) protocol to send important notifications to the healthcare facilities/users in places with low or no internet connection.
  - ✓ Purpose: To have a reliable fallback mechanism of urgent notice (e.g. request received, critical low stock alert) in areas where data connectivity is not reliable. It is an optional feature that is concerned with a third-party SMS Gateway.
  - ✓ Connection Specifics: Exchange with a third-party SMS Gateway API (e.g. through an HTTPS POST request to the endpoint of the gateway). The underlying SMS protocol will then be taken care of by the gateway. The content of messages will be brief and pre-made.
  - ✓ Reference: To be integrated Specific SMS Gateway API documentation.
- VBB-COMM-003: The system will make use of such push notifications as Firebase Cloud Messaging (FCM) (of Android) or Apple Push Notification service (APNs) (of iOS).
  - ✓ Purpose: To send real-time and targeted messages to mobile devices, even when the app is not currently in use, to notify interests such as new blood requests or updates to their status.
  - ✓ Protocol Description: The backend (VBB-BACKEND-API) will be connected to FCM/APNs destinations to transmit the messages, which will then take care of the proprietary push notification protocols to deliver the messages to the corresponding mobile clients.
  - ✓ Source: Firebase Cloud messaging documentation, Apple push notification service documentation.

## 3.2 Functional Requirements

This section outlines the particular functions that the Virtual Blood Bank mobile application will be playing. These requirements specify what the system will do in response to the input and the output it will give, in an externally observable view. All requirements are designated, verifiable and not specific to design implementation.

### 3.2.1. Feature or Functional Requirement 1.

#### **3.2.1 Blood Inventory Management**

The feature characterizes the ability of the system to allow healthcare facilities to manage and see their stock levels and associated information of the blood.

##### **3.2.1.1 Introduction**

This is a list of specifications that describe the way that the healthcare workers will be able to see, update, and track the blood inventory of the facilities they are working with and the other facilities in their district. This is essential to facilitate visibility in real time and effective resource allocation.

##### **3.2.1.2 Inputs**

- VBB-IN-BIM-001: Inventory is updated when the user clicks on the update inventory option.
- VBB-IN-BIM-002: User key Blood Type (enum: A+, A-, B+, B-, AB+, AB-, O+, O-).
- VBB-IN-BIM-003: Number of Units (int, 1-999).
- VBB-IN-BIM-004: Expiration Date (date, YYYY-MM-DD, should be future date) is entered by the user.
- VBB-IN-BIM-005: Selects submit to update inventory.
- VBB-IN-BIM-006: User chooses the option View My Facility Inventory.
- VBB-IN-BIM-007: "View District Inventory" is selected by user.
- VBB-IN-BIM-008: Query Search Query (string, e.g., facility name, type of blood to filter out district view).

##### **3.2.1.3 Processing**

- VBB-PROC-BIM-001: The system will authenticate all inputs that are related to inventory update (Blood Type, Number of Units, Expiration Date).
- VBB-PROC-BIM-002: In case of successful validation and submission the system will update the inventory records of the user/facility being related to the central database.
- VBB-PROC-BIM-003: The system will acquire the existing data on blood inventory within the associated facility of the user in the central database.
- VBB-PROC-BIM-004: The system will access the current data of blood inventory at all the cooperating facilities within the district of the user in the central database.
- VBB-PROC-BIM-005: The system will be able to filter and sort data related to the district inventory according to user search query or other criteria of choice.
- VBB-PROC-BIM-006: The system will be configured to alert blood units that are about to expire (within 7 days) in the inventory displays.
- VBB-PROC-BIM-007: The system will be configured to provide warning on facilities with critically low stock levels (configurable per blood type, e.g., less than 2 units).

##### **3.2.1.4 Outputs**

- VBB-OUT-BIM-001: The current stock of blood inventory is shown to the user of his/her facility: each record contains the Blood Type, the Number of Units, and the Expiration Date.

- VBB-OUT-BIM-002: A consolidated dashboard with blood inventory summary on all cooperating facilities in the district with the following information: Facility Name, Blood Type, and Number of Units.
- VBB-OUT-BIM-003: Visual cues (e.g., color-coded, icon) of facilities that have critically low stock volume or where blood units are expiring.
- VBB-OUT-BIM-004: Message of confirmation of successful inventory update.
- VBB-OUT-BIM-005: List of filtered / sorted list of facilities and their blood stock on the search input by the user.

### 3.2.1.5 Error Handling

- VBB-ERR-BIM-001: The system will be shown with an error message in case the necessary inventory update fields are not found or are not valid (e.g. non-numeric unit, expiration date is in the past, etc.).
- VBB-ERR-BIM-002: The system will show an error notice in case of failure to update inventory because of network connection problems, which would prompt the user to re-attempt to update the inventory.
- VBB-ERR-BIM-003: In case no inventory data is available to view any information about any facility or the district, the system will show a message of no data available.

## 3.2.2 Blood Requests and Tracking.

This feature outlines the capabilities of the system as far as a healthcare professional is concerned in terms of locating and ordering blood in another facility and monitoring the status of their pending orders.

### 3.2.2.1 Introduction

The following list of requirements describes the fundamental emergency purpose of the application: allowing a user at the facility in need to find and officially request blood in a cooperating facility with an excess. It also addresses the way the user could keep track of the status of his requests.

### 3.2.2.2 Inputs

- VBB-IN-BRQ-001: User has made an option of Find Blood or District Inventory.
- VBB-IN-BRQ-002: The filter criteria are typed by the user (e.g. Blood Type).
- VBB-IN-BRQ-003: User will choose a target providing facility in the list/map provided.
- VBB-IN-BRQ-004: Number of Units required (int, positive).
- VBB-IN-BRQ-005: User can add optional notes to the request (text, up to 500 chars).
- VBB-IN-BRQ-006: The user clicks on Submit Request.
- VBB-IN-BRQ-007: User clicks to his/her Outgoing Requests screen.

### 3.2.2.3 Processing

- VBB-PROC-BRQ-001: The system will filter the list of district facilities according to the blood type that is specified by the user and show only the ones that have the units available.
- VBB-PROC-BRQ-002: The system will check the request entries when they are submitted.
- VBB-PROC-BRQ-003: The system will ensure that the ordered quantity of units requested does not exceed the available quantity of units at the preferred supplying facility.
- VBB-PROC-BRQ-004: Once the validation is successful the system will generate a new record of a blood request in the database, which will have a starting status of a pending request.
- VBB-PROC-BRQ-005: The system will automatically send a notification to the relevant users at the chosen facility of supply.
- VBB-PROC-BRQ-006: The system will be able to retrieve and show all the blood requests that have been issued by the facility of the user, their status, and the most important information.



#### 3.2.2.4 Outputs

- VBB-OUT-BRQ-001: Presentation of a list or map of available blood facilities with Facility Name, District and Units Available of the given type of blood.
- VBB-OUT-BRQ-002: There was a pre-filled request form with the name of the chosen facility.
- VBB-OUT-BRQ-003: A party confirmation of a successful process when a blood request is submitted.
- VBB-OUT-BRQ-004: A list of outgoing Requests on a screen with the following fields: RequestID, SupplyingFacility, BloodType, UnitsRequested, Status, and Timestamp.

#### 3.2.2.5 Error Handling

- VBB-ERR-BRQ-001: In case there are not any facilities with the requested type of blood, the system must show a message saying that there are no stocks in the district.
- VBB-ERR-BRQ-002: In case the user requests in excess of the available units, the system will never allow it but will notify the user to have an error message.
- VBB-ERR-BRQ-003: In case of an error during the request submission because of a network error, the system should show an error message, thus offering a retry option.

### 3.2.3 Incoming Blood Request Management.

This characteristic explains the possibility of the medical professional to see and act on the blood demands posted by other institutions.

#### 3.2.3.1 Introduction

This can be described as the opposite of requesting feature. It enables a facility that has an excess of blood to handle requests that may come in either accepting them to offer help or declining them in case they are not able to supply the request.

#### 3.2.3.2 Inputs

- VBB-IN-IBR-001: User becomes familiar with the screen of Incoming Requests.
- VBB-IN-IBR-002: User will choose a particular request on the list.
- VBB-IN-IBR-003: User chooses an action: Accept request or reject request.
- VBB-IN-IBR-004: User is offered to provide an optional reason of rejection (text, 255 chars).
- VBB-IN-IBR-005: User changes the status of an accepted request (e.g., clicks on "Mark as In Transit, Mark as Fulfilled).

#### 3.2.3.3 Processing

- VBB-PROC-IBR-001: The system will store and display a list of all pending blood orders that have been dispatched to the facility of the user.
- VBB-PROC-IBR-002: When the user clicks on Accept Request: the system will:
  - ✓ Flip the request status in the data base to accepted.
  - ✓ Reduce the available inventory (or star out the units) to avoid the possibility of the same units being promised to more than one requester.
  - ✓ Send a notification to the original requesting facility of the update in the status.
- VBB-PROC-IBR-003: When the user clicks on the option of Reject Request, the system will:
  - ✓ Set the request database to the status of rejected.
  - ✓ Save the cause of rejection, where given.
  - ✓ Send a notification to the sending requesting facility with the status update.
- VBB-PROC-IBR-004: An authorized user shall have the ability to change the status of accepted request using the lifecycle of such request by its status (In Transit, Fulfilled) and send notifications when statuses are changed.

#### 3.2.3.4 Outputs

- VBB- OUT-IBR-001: A list of incoming requests on a screen with the fields of RequestID, RequestedFacility, BloodType, UnitsRequested, Status, and Timestamp.
- VBB-OUT-IBR-002: The details of a chosen request.
- VBB-OUT-IBR-003: A confirmation message upon the successful acceptance of a request, or a rejection of a request.

#### **3.2.3.5 Error Handling**

- VBB-ERR-IBR-001: In case a user tries to accept a request but his/her facility inventory has decreased to less than the required amount since the last request, the system will also show an Insufficient inventory to fulfill request error, and deny acceptance.
- VBB-ERR-IBR-002: In case status update fails because of a network error, the system will require an error message - and give an option to re-try.

### **3.2.4 Donor Center Mapping**

This feature explains how the system is able to show users information regarding the location of blood donors centers within a certain area.

#### **3.2.4.1 Introduction**

This feature will ensure that the blood supply is sustainable by providing easy access to the information on where to give blood by the users (and by extension, the community).

#### **3.2.4.2 Inputs**

- VBB-IN-DCM-001: The choice of the menu is to select the option of Find Donation Centers.
- VBB-IN-DCM-002: User authorizes the application to get the current location of the device.

#### **3.2.4.3 Processing**

- VBB-PROC-DCM-001: A list of all blood donation centers registered on the database shall be retrieved by the system.
- VBB-PROC-DCM-002: In case location permission is allowed, then the system will find out the current geographical location of the user.
- VBB-PROC-DCM-003: The system will include donation centers on an interactive map which will have pins placed on top of them.
- VBB-PROC-DCM-004: The system will give a list view of the donation centers, which can be either ordered by proximity (where possible) by the current position of the user or alphabetically.

#### **3.2.4.4 Outputs**

- VBB-OUT-DCM-001: an interactive map with the points of blood donation centres.
- VBB-OUT-DCM-002: List of blood donation centers that can be scrolled.
- VBB-OUT-DCM-003: Each center in the model represents one detail, in which the name, address, hours of operation, and contact information are displayed.

#### **3.2.4.5 Error Handling**

- VBB-ERR-DCM-001: In case of denial of location permission, the system will show the list of all the centers in alphabetical order but not sorted by distance and show a message stating that location is required to sort proximity.
- VBB-ERR-DCM-002: In case no donation centers are registered in the system, then the system will display a message indicating that there are no donation centers registered.



### 3.3 Use Cases

This section describes the detailed, user-centric interactions with the system to accomplish key goals. Each use case is presented in a formal structure, detailing the actor, goal, scenarios, and extensions from the actor's point of view.

#### 3.3.1 Use Case: Blood Inventory Update Facility

| Field                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID                    | VBB-UC-001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Use Case Name                  | Update Facility Blood Inventory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Goal                           | To make the records of blood stock at the facility of a Healthcare Professional up to date to show the current physical inventory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Primary Actor                  | Healthcare Professional (e.g., Nurse, Hospital Administrator, Blood Bank Technician)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Precondition                   | The Healthcare Professional is a registered and logged-in user to the application.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Trigger                        | The physical blood inventory in the facility is modified (e.g. a new unit arrives, a unit is utilized), and the Healthcare Professional triggers the change process.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Main Success Scenario          | <ol style="list-style-type: none"> <li>1. The Healthcare Professional specifies that he/she would like to update blood inventory at his/her facility.</li> <li>2. The system shows the existing inventory and allows one to edit it.</li> <li>3. The Healthcare Professional identifies the blood type, new amount of units (or alteration) and the date on which any new units expire.</li> <li>4. The changes are confirmed by the Healthcare Professional.</li> <li>5. The system authenticates the information that is entered into it.</li> <li>6. The system takes the update and aligns it with the central server.</li> <li>7. The system makes the Healthcare Professional sure that the inventory is now up-to-date.</li> </ol> |
| Success End Condition          | The blood inventory in the facility is properly updated in the system and is accessible to all users of the district.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Failure End Condition          | The inventory record of the system has not been changed, or it has not been synchronized with the central server.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Extensions (Failure Scenarios) | <p><b>5. The data provided is invalid (e.g. quantity is not a number, expiration date is past):</b></p> <ul style="list-style-type: none"> <li>✓ 5.1. The system declines update and provides the Healthcare Professional that there is a certain error.</li> <li>✓ 5.2. The Healthcare Professional fixes the data and goes back to the step 4.</li> </ul> <p><b>6. System cannot synchronise the update with the central server because of a network</b></p>                                                                                                                                                                                                                                                                            |

|  |                                                                                                                                                                                                                                                                                                                                                  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>failure:</b> <ul style="list-style-type: none"> <li>✓ 6.1. The Healthcare Professional is notified about the connectivity problem by the system.</li> <li>✓ 6.2. The system logs the update on the local machine and will notify the user that it shall automatically make an effort to synchronise upon recovery of connectivity.</li> </ul> |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 3.3.2 Use Case: Request Blood from Another Facility

| Field                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID                    | VBB-UC-002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Use Case Name                  | Request Blood in Another Facility.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Purpose                        | To seek and demand emergency blood in a nearby facility by a Healthcare Professional.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Primary Actor                  | Requesting Medical Personnel (e.g., Doctor, Nurse)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Supporting Actor               | Supplying Healthcare Professional (system notified)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Precondition                   | The Requested Healthcare Professional is identified and has recognized an emergency patient requirement of a particular type of blood. The local inventory of the facility is not enough.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Trigger                        | A patient has an urgent need of blood transfusion, but the local stock in the facility is not enough.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Main Success Scenario          | <ol style="list-style-type: none"> <li>1. Requesting Healthcare Professional launches a search of a given type of blood in the district.</li> <li>2. The system shows a list of facilities that have stocked the required type of blood and the quantity of each.</li> <li>3. Requesting Healthcare Professional chooses an appropriate supplying facility in the list.</li> <li>4. Requesting Healthcare Professional indicates the quantity of units to be ordered and confirms the request.</li> <li>5. The system ensures that the requested quantity is on-hand at the supplying plant.</li> <li>6. This system logs down the request, and assigns it a status of pending and right away sends out a notification to the assigned users of the supplying facility.</li> <li>7. The system sends feedback to the Requesting Healthcare professional that the request has been relayed successfully.</li> </ol> |
| Success End Condition          | A formal blood request is sent to the supplying facility successfully, and the Requesting Healthcare Professional is confirmed that it is pending.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Failure End Condition          | No request is forwarded, or a request cannot be generated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Extensions (Failure Scenarios) | <b>2. In the district, System has not discovered facilities of the required type of blood:</b> <ul style="list-style-type: none"> <li>✓ 2.1. The system lets the Requested Healthcare Professional know that there is no inventory at the moment.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>✓ 2.2. The use case ends.</p> <p><b>5. The quantity ordered is higher than the inventory can be provided by the supplying facility (e.g., some order has just been completed):</b></p> <p>✓ 5.1. The system declines the request submission and notifies the professional on the available quantity at a particular time.</p> <p>✓ 5.2. The Requesting Healthcare Professional may prefer to realign the quantity (goes back to step 4) or option to change facilities (goes back to step 3).</p> <p><b>6. Request cannot be sent to the system because of a network failure:</b></p> <p>✓ 6.1. The system notifies the professional of the connectivity problem and gives him an option of resubmitting (goes back to step 4).</p> |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 3.3.3 Use Case: Respond to an Incoming Blood Request

| Field                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>           | VBB-UC-003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Use Case Name</b>         | Incoming Blood Request Response.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Purpose</b>               | To review and reply to a blood request made by another facility to a Healthcare Professional.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Primary Actor</b>         | Providing Healthcare Profession (e.g., Nurse, Administrator)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Precondition</b>          | The Supplying Healthcare Professional is authenticated, and their facility is issued with a pending blood request.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Trigger</b>               | The Supplying Healthcare Professional receives an incoming request notification of a new blood request.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Main Success Scenario</b> | <ol style="list-style-type: none"> <li>1. The Supplying Healthcare Professional perceives the list of incoming requests.</li> <li>2. The Supplying Healthcare Professional picks a request in progress so as to examine its contents.</li> <li>3. The system shows the information about the request (requesting facility, blood type, units, etc.).</li> <li>4. The Supplying Healthcare Professional authenticates that they have the stock and makes a choice to accept the request.</li> <li>5. The system changes the request status to the one of the Accepted.</li> <li>6. The system will reduce the available inventory of the facility by the required quantity of that type of blood.</li> <li>7. The system makes a notification to the requesting facility on acceptance.</li> <li>8. The Supplying Healthcare Professional is provided with a confirmation of the action by the</li> </ol> |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Success End Condition</b>          | The received request is either Accepted and the inventory is changed, or it is Rejected and the requesting facility is informed by the choice.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Failure End Condition</b>          | The request status is left at "Pending" status.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Extensions (Failure Scenarios)</b> | <p><b>4. The Supplying Healthcare Professional resolves to decline the request:</b></p> <ul style="list-style-type: none"> <li>✓ 4.1. The Supplying Healthcare Professional picks off Reject.</li> <li>✓ 4.2. The system changes the request status to the rejection status.</li> <li>✓ 4.3. The system notifies the requesting facility on the rejection.</li> <li>✓ 4.4. The use case is successfully completed.</li> </ul> <p><b>6. The system then concludes that the physical inventory is no longer adequate to meet the request (e.g. a unit was consumed locally immediately before being accepted):</b></p> <ul style="list-style-type: none"> <li>✓ 6.1. The system does not allow the action Accept.</li> <li>✓ 6.2. This system notifies the Supplying Healthcare Professional that there is no longer enough stock and recommends that they deny the request or call the requester.</li> </ul> |

### 3.3.5 Use Case: Track Status of Blood Requests

| Field                        | Description                                                                                                                                        |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>           | VBB-UC-005                                                                                                                                         |
| <b>Goal</b>                  | A Healthcare Professional wishes to monitor the current status of their facility's pending and active blood requests (both outgoing and incoming). |
| <b>Primary Actor</b>         | Healthcare Professional (Requesting or Supplying)                                                                                                  |
| <b>Scope</b>                 | Virtual Blood Bank System                                                                                                                          |
| <b>Level</b>                 | User Goal                                                                                                                                          |
| <b>Precondition</b>          | The Healthcare Professional is authenticated, and their facility has either sent or received at least one blood request.                           |
| <b>Success End Condition</b> | The user is presented with an up-to-date list of relevant requests and their current statuses.                                                     |
| <b>Failure End Condition</b> | The user is unable to retrieve the status of their requests.                                                                                       |
| <b>Trigger</b>               | The user needs to know the progress of a blood request to make a clinical or logistical decision.                                                  |
| <b>Main Success</b>          | 1. Healthcare Professional indicates their intent to view blood requests (e.g., selects "Outgoing                                                  |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scenario</b>                       | <p>Requests" or "Incoming Requests").</p> <p>2. System retrieves all relevant requests associated with the user's facility from the database.</p> <p>3. System displays a summary list of the requests, showing key information for each: a unique ID, the name of the other facility involved, the blood type, and the current Status (e.g., "Pending," "Accepted," "Rejected," "In Transit," "Fulfilled").</p> <p>4. Healthcare Professional selects a specific request from the list to see more details.</p> <p>5. System displays a detailed view of the selected request, including a history of status changes with timestamps.</p>                                                                                                                                   |
| <b>Extensions (Failure Scenarios)</b> | <p><b>3a.</b> System finds no requests for the user's facility.</p> <p>✓ <b>3a.1.</b> System displays a clear message to the user, such as "No outgoing requests found."</p> <p>✓ <b>3a.2.</b> The use case ends successfully.</p> <p>✓ <b>3b.</b> System is unable to retrieve the request list due to a network failure.</p> <p>✓ <b>3b.</b> System is unable to retrieve the request list due to a network failure.</p> <p>✓ <b>3b.1.</b> System informs the user of the connectivity issue.</p> <p>✓ <b>3b.2.</b> System displays a cached version of the request list from the last successful synchronization, with a clear warning that the information may be outdated.</p> <p>✓ <b>3b.3.</b> User can choose to manually refresh once connectivity is restored.</p> |

### 3.3.6 Use Case: Administer User Accounts

| Field                        | Description                                                                                                                   |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>           | VBB-UC-006                                                                                                                    |
| <b>Goal</b>                  | An administrator needs to create, deactivate, or manage user accounts for their designated facility or district.              |
| <b>Primary Actor</b>         | An administrator needs to create, deactivate, or manage user accounts for their designated facility or district.              |
| <b>Scope</b>                 | Virtual Blood Bank System                                                                                                     |
| <b>Level</b>                 | User Goal                                                                                                                     |
| <b>Precondition</b>          | The Facility Administrator is authenticated and logged into the application with administrative privileges.                   |
| <b>Success End Condition</b> | A new user account is successfully created and the user is notified, OR an existing user account is successfully deactivated. |
| <b>Failure End Condition</b> | The user account list remains unchanged.                                                                                      |
| <b>Trigger</b>               | A new healthcare professional needs access to the system, or an existing user's access needs to be revoked.                   |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Main Success Scenario</b>          | <ol style="list-style-type: none"> <li>1. Facility Administrator navigates to the "User Administration" section.</li> <li>2. System displays a list of current users for the administrator's facility/district.</li> <li>3. Administrator selects the option to "Create New User."</li> <li>4. System presents a form for new user details.</li> <li>5. Administrator enters the required information: Full Name, Phone Number, Email Address (optional), associated Facility, and Role (e.g., Doctor, Nurse).</li> <li>6. Administrator confirms the submission.</li> <li>7. System validates that the inputs are complete and that the Phone Number/Email is unique.</li> <li>8. System creates the new user account in the database, generates a temporary password, and marks the account as active.</li> <li>9. System sends the new user their login credentials (e.g., via SMS to their registered phone number).</li> </ol> <p>System confirms to the administrator that the account has been created successfully.</p> <p><u>Variations (Deactivating an Existing User):</u></p> <ol style="list-style-type: none"> <li>3'. Administrator selects an existing user from the list displayed in step 2.</li> <li>4'. Administrator chooses the "Deactivate User" option.</li> <li>5'. System prompts for confirmation to prevent accidental deactivation.</li> <li>6'. Administrator confirms the action.</li> <li>7'. System updates the user's account status to "Inactive," which will prevent them from logging in.</li> <li>8'. System confirms to the administrator that the account has been deactivated.</li> </ol> |
| <b>Extensions (Failure Scenarios)</b> | <ul style="list-style-type: none"> <li>✓ 6a'. Administrator attempts to deactivate their own account.</li> <li>✓ 6a'.1. System prevents the action and displays a warning message, such as "Administrators cannot deactivate their own account."</li> <li>✓ 7a. Submitted data is invalid or not unique.</li> <li>✓ 7a.1. System rejects the submission and displays an error message to the administrator indicating which field is invalid (e.g., "Phone number already in use.").</li> <li>✓ 7a.2. The administrator corrects the data and returns to step 6.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### 3.3.7 Use Case: Log into the System

| Field                        | Description                                                                                                               |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>           | VBB-UC-007                                                                                                                |
| <b>Goal</b>                  | A user wishes to authenticate themselves to gain secure access to the application's features.                             |
| <b>Primary Actor</b>         | Healthcare Professional (includes all roles, including Administrator)                                                     |
| <b>Scope</b>                 | Virtual Blood Bank System                                                                                                 |
| <b>Level</b>                 | Subfunction                                                                                                               |
| <b>Precondition</b>          | The user has an active, registered account and is viewing the application's login screen.                                 |
| <b>Success End Condition</b> | The user is successfully authenticated, a secure session is established, and the main application dashboard is displayed. |
| <b>Failure End Condition</b> | The user is denied access and remains on the login screen.                                                                |
| <b>Trigger</b>               | The user opens the application or their previous session has expired.                                                     |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Main Success Scenario</b>          | <ol style="list-style-type: none"> <li>1. User enters their registered credentials (Phone Number or Email Address) and their password.</li> <li>2. User indicates their intent to log in.</li> <li>3. System securely transmits the credentials to the backend for verification.</li> <li>4. System validates that the credentials are correct and that the user's account is marked as "Active."</li> <li>5. System establishes a secure session and presents the main application dashboard.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Extensions Scenarios) (Failure</b> | <ol style="list-style-type: none"> <li>1a. User has forgotten their password.               <ol style="list-style-type: none"> <li>1a.1. User selects the "Forgot Password" option.</li> <li>1a.2. System initiates the password reset workflow (this is a separate use case).</li> </ol> </li> <li>4a. The submitted credentials are incorrect.               <ol style="list-style-type: none"> <li>4a.1. System rejects the authentication attempt.</li> <li>4a.2. System displays a generic error message, such as "Invalid credentials. Please try again."</li> <li>4a.3. User is returned to step 1.</li> </ol> </li> <li>4b. The user's account is "Inactive" or "Deactivated."               <ol style="list-style-type: none"> <li>4b.1. System rejects the authentication attempt.</li> <li>4b.2. System displays a specific message, such as "Your account has been deactivated. Please contact your administrator."</li> <li>4b.3. The use case ends in failure.</li> </ol> </li> <li>4c. The user's account is temporarily locked (e.g., after too many failed login attempts).               <ol style="list-style-type: none"> <li>4c.1. System rejects the authentication attempt.</li> <li>4c.2. System displays a message, such as "This account is temporarily locked due to too many failed login attempts. Please try again later or reset your password."</li> <li>4c.3. The use case ends in failure.</li> </ol> </li> </ol> |

**Note: Due to page layout constraints and to ensure the best visibility of the use case diagram, it has been moved to the final page.**

### 3.4 Non-Functional Requirements

This section would outline the non-functional requirements (NFRs) of the Virtual Blood Bank mobile application. These requirements include quality attributes of the system which are vital to the overall success of the system, and not particular features. They are generally systemic, are usually expressed in terms of measure to be verifiable.

#### 3.4.1 Performance

- VBB-NFR-PERF-001: The system will show the user blood inventory information of his or her facility and the district within 5 seconds after a request is made under normal network conditions (e.g. 3G/4G connection).
- VBB-NFR-PERF-002: The system will respond to blood inventory updates (add/modify units) in less than 3 seconds of the mobile application submission in conditions of normal network performance.



- VBB-NFR-PERF-003: The system will provide users with the most important in-app notifications (e.g., new blood request, low stock alert) in under 60 seconds after the event on the backend has happened.
- VBB-NFR-PERF-004: The system will take a maximum of 3 seconds to respond to users who are attempting to log in under normal network conditions.
- VBB-NFR-PERF-005: The mobile application shall not use over 50 MB of background RAM on an average of more than 5 minutes when idle.
- VBB-NFR-PERF-006: The mobile application will finish a data synchronization (e.g., list of facilities, initial blood inventory of own facility) within 15 seconds when it is first launched after being installed or a major update is performed provided that the network connection is stable.

#### 3.4.2 Reliability

- VBB-NFR-REL-001: The system will maintain integrity of blood inventory record and the data corruption rate will not exceed 0.01 of the total blood inventory records per quarter.
- VBB-NFR-REL-002: The system will be able to effectively fulfill the 99.9 percent of all blood request transactions (creation, update, cancellation) with no loss of data or errors instigated by the system.
- VBB-NFR-REL-003: The system will not lose unexpectedly terminated power or application crashes without submitting (and not yet synchronized) critical user data (e.g. pending inventory updates) within 5 seconds of application re-initiation.
- VBB-NFR-REL-004: The Mean Time Between Failures (MTBF) of the important backend services (e.g. database, API server) should be more than 30 days.
- VBB-NFR-REL-005: The system will make sure that all crucial notifications are received by the device of the recipient with a success rate of 98 and above (excluding factors that are outside the control of the system, e.g. a device being offline or notifications are turned off by the user).

#### 3.4.3 Availability

- VBB- NFR- AVAIL-001: The Virtual Blood Bank backend services will be provided 99.5 percent of the time (but not during planned maintenance windows of not more than 4 hours a month).
- VBB-NFR-AVAIL-002: The mobile application will be available and operational on the supported devices even when accessing intermittent or no internet connection to view the data stored in the cache and compose the requests offline. When connection is reestablished, data synchronization will be carried out automatically.
- VBB-NFR-AVAIL-003: The maintenance or update time of the planned system downtime will not exceed 4 hours a month and it will be planned as occurring during off-peak hours (e.g., between 12:00 AM and 4:00 AM local time).
- VBB-NFR-AVAIL-004: The system will allow a minimum of 24 hours of access to the cache data of blood inventory when offline to last synchronization, which was successful.

#### 3.4.4 Security

- VBB-NFR-SEC-001: The system will encrypt all sensitive information (e.g., passwords, personally identifiable information, blood inventory data) during transmission (via TLS 1.2 or more) and storage (via AES-256 or higher encryption).

- VBB-NFR-SEC-002: The system will include role-based access control (RBAC), whereby the users should be able to access only those functions and data relevant to their designated role and relevant facility/district.
- VBB-NFR-SEC-003: The system will not be vulnerable to SQL injections, cross-site scripting (XSS), as well as other typical internet vulnerabilities according to the OWASP Top 10.
- VBB-NFR-SEC-004: The system will enforce the user passwords to have a minimum complexity level (e.g., 8 characters, including uppercase, lowercase, digits, and special characters) and it will store the passwords with strong hashing algorithms (e.g. bcrypt, PBKDF2).
- VBB-NFR-SEC-005: An inactive user would automatically be logged out within 30 minutes (as per VBB-FUN-UM-005).
- VBB-NFR-SEC-006: The system will have an audit trail of all important activities (e.g., blood inventory updates, blood request generated/approved) that will capture the user, activity, time, and data impacted. Such logs will also be maintained at least 1 year.
- VBB-NFR-SEC-007: The system will not allow unauthorized users to access the backend API: All data manipulation operations will be governed by a valid authentication token.

### 3.4.5 Maintainability

- VBB-NFR-MAINT-001: The codebase of the system will comply with the accepted code standards and will be covered with detailed internal documentation, and the code maintainability index score will be at least 75% (with the help of such tools as SonarQube or similar).
- VBB-NFR-MAINT-002: The system will have a modular design which will enable the modification or replacement of components without affecting other unrelated parts of the system at 80 percent of the modules identified.
- VBB-NFR-MAINT-003: The system will support the implementation of software updates (patches, new features) to production environments with an average of less than 2 hours to rollout and test and verify.
- VBB-NFR-MAINT-004: The system will offer easily defined error records of the back-end services, which will allow the detection and troubleshooting of problems within 30 minutes after they have happened.

### 3.4.6 Portability

- VBB-NFR-PORT-001: The mobile application is fully compatible with Android devices and operating Android 8.0 (Oreo) and later.
- VBB-NFR-PORT-002: The mobile application must be able to be launched and work in full on iOS devices that are running iOS 13.0 or newer, and can be cross-platform based (e.g., React Native or Flutter).
- VBB-NFR-PORT-003: The backend services will be designed so that they can be deployed on cloud environments (e.g., GCP, AWS) with the use of containerization technologies (e.g., Docker) to allow environment independency and easiness of moving to the new environment.
- VBB-NFR-PORT-004: The database schema of the system will be such that it can support most of the relational (e.g., PostgreSQL) or NoSQL (e.g., MongoDB) database systems with ease without any major refactoring.

## 3.5 Inverse Requirements

Inverse requirements clearly understand what the system will not do or what it will not take care of. They can be applied to establish boundaries, avoid scope creep and make sure that stakeholders and developers understand each other.

- VBB-INV-001: The system will neither create nor operate or fund new physical blood bank facilities and blood storage units. Any physical blood storage is the prerogative of the current healthcare facilities.
- VBB-INV-002: The system will not offer or administer any tangible transportation of blood logistics and infrastructure (e.g., vehicles, drones, specific people in charge of transport). The requesting and supplying facilities still have the responsibility of coordination of physical transport between facilities.
- VBB-INV-003: It will not do any medical blood analyticals, compatibility or infectious disease screening, blood analyticals (e.g., component separation) or quality control. Such roles only belong to the mandate of relevant healthcare workers and national blood services (e.g., Ethiopian Blood and Tissue Bank Service).
- VBB-INV-004: The system will not be collecting blood donations directly out of the individuals. It is restricted to giving information about the available donation centers through a mapping facility about its involvement with donors.
- VBB-INV-005: The system will not give medical diagnosis or treatment recommendations, as well as instructions on patient care. Its role is to deliver information and manage the resources.
- VBB-INV-006: The system will not ensure the presence of blood at all times, but will give real-time information on what is available currently in the participating facilities.
- VBB-INV-007: The system will not interface or seek to alter the functionality of the already present Ethiopian Blood Safety Information System (BSIS) or any other national-level health information systems without future requirement and distinct interface specifications.

### 3.6 Design Constraints

This sub section outlines certain restrictions under the influences of external standard, company policies, regulatory requirements, or other factors, which cannot be negotiable and put direct restrictions on the options the developer can make in designing the software system. These are more prescriptive than general constraints in Section 2.4, and directly have an influence on technical design decisions.

- VBB-DC-001: The mobile app of the system will be built on cross-platform architecture (e.g., React Native or Flutter) to run on Android and possibly on iOS platform with a single codebase due to the cost-efficiency and resource optimization.
- VBB-DC-002: Python with Django framework and Django Rest Framework will be used as the backend service to the system based on the available team experience and fast development features.
- VBB-DC-003: The database solution within the system is to be a database cloud (e.g., MongoDB or PostgreSQL on GCP/AWS) to ensure that the local infrastructure can be kept to the minimum and the scalability and managed services can be used. The decision between MongoDB and PostgreSQL will be a design decision, yet it has to be of this kind.
- VBB-DC-004: The mobile client and the backend should encrypt all data transmitted using TLS 1.2 or above, which is a requirement of security policies and regulatory requirements of health data.

- VBB-DC-005: The system will be deployed on a significant cloud system (Google Cloud platform (GCP) or Amazon Web Services (AWS)) to be based on the quality, scalability, and cost-effectiveness of the solution based on the project financing and infrastructure plan.
- VBB-DC-006: The system will comply with the local Ethiopian laws on data residency of health information, that is, all primary data storage of the backend services should be within the geographical boundaries of the country in case of their being imposed by the laws of the country.
- VBB-DC-007: The mobile application UI/UX design must have a mobile-first design that is optimized to smaller screens and touch interaction in order to accommodate tablet or larger screen sizes.

### 3.7 Database Requirements Logical.

This sub section provides the logical requirements of the system database, such as data formats, storage, data retention and integrity but it does not specify the implementation physically, only restrictions as specified by 3.6.

- VBB-DB-001: The data concerning Healthcares will be stored in the database in terms of:
  - ✓ FacilityID (Unique Identifier, Primary Key)
  - ✓ Name (Text, Max 255 chars)
  - ✓ District (Text, Max 100 chars)
  - ✓ Location (Geospatial coordinates: Posited, Longitude)
  - ✓ ContactPhoneNumber (Text, up to 20 chars)
  - ✓ EmailAddress (Text, Max 255 chars, Optional)
  - ✓ IsWorking (Default True, Boolean, Flag)
- VBB-DB-002: Data of Users will be included in the database, including:
  - ✓ UserID (Unique Identifier, PK)
  - ✓ FacilityID (Foreign Key of Facilities)
  - ✓ FullName (Text, Max 100 chars)
  - ✓ Text (email, Maximum 255 characters, Unique, Optional, used to log in)
  - ✓ PhoneNumber (Text, Max 20 chars, Uniqueness, used as a login/contact)
  - ✓ PasswordHash (Text, Stores hashed password)
  - ✓ Role (Enum: Doctor, Nurse, Administrator, Blood Bank Technician)
  - ✓ IsActive (Default True, Boolean): A Boolean.
  - ✓ LastLogin (Timestamp)
- VBB-DB-003: The Database in every given facility will have the Blood Inventory Records that will include:
  - ✓ InventoryitemID(Unique identifier, PK)
  - ✓ FacilityID (Foreign Key to facilities)
  - ✓ BloodType (Enum: A+, A-, B+, B-, AB+, AB-, O+, O-)
  - ✓ There is Availableunits (Integer, Non-negative): This is the units available in a product.
  - ✓ ExpirationDate (Date)
  - ✓ LastUpdated (Timestamp)
- VBB-DB-004: Blood Request Records This shall be stored in the database and shall include:
  - ✓ RequestID (Primary Key, Unique Identifier)
  - ✓ RequestingfacilityID (Foreign key to facilities)
  - ✓ Facilities (Number)

- ✓ SupplyingFacilityID (Null but not unique foreign key of Facilities)
- ✓ RequestedBloodType (A+, A-, B+, B-, AB+, AB-, O+, O -)
- ✓ Requested (Integer, Positive)
- ✓ RequestTimestamp (Timestamp)
- ✓ Status (Enum: Pending, Accepted, Rejected, In Transit, Fulfilled, Canceled)
- ✓ AcceptanceTimestamp (Timestamp, Nullable)
- ✓ Timestamp (Timestamp, Nullable) DispatchTimestamp
- ✓ FulfillmentTimestamp (Timestamp, Nullable)
- ✓ Notes (Text, Max 500 chars, Optional)
- VBB-DB-005: The information in the database will be the Blood Donation Center Information, which will include:
  - ✓ DepictionID (Unique Identifier, Foreign Key)
  - ✓ Name (Text, Max 255 chars)
  - ✓ District (Text, Max 100 chars)
  - ✓ Location (Geospatial coordinates: Latitude, Longitude)
  - ✓ OperatingHours (Text, e.g., "Mon-Fri 9AM-5PM")
  - ✓ ContactInfo (Text, Max 255 chars)
- VBB-DB-006: There will be a referential integrity between related entities (i.e. FacilityID in Users should refer to an existing FacilityID in Facilities).
- VBB-DB-007: The database will ensure the data consistency (all the changes to the data (additions, subtraction to the BloodType and FacilityID of the inventory), i.e. any changes to the UnitsAvailable will be linked directly to the BBB changes to the BloodType (addition) or a FacilityID (subtraction).
- VBB-DB-008: The database will have a capacity of storing 500 or more healthcare centers and 5000 or more user accounts within a district, and can support any type of district.
- VBB-DB-009: The database should store all the audit trail records (as per VBB-NFR-SEC-006) in at least 1 year. More data on operations (e.g. fulfilled blood requests) will be kept at least 3 years to study the history.

### 3.8 Other Requirements

*This segment attaches any other requirements that do not comfortably fall under the functional, non-functional or interface category but are critical to the success of the project and the eventual implementation and usage of the software.*

*Training requirements are related to training requirements.*

- VBB-TRN-001: *The system will be provided with a brief and user-friendly digital user manual that will be available instantly on the mobile application. All the fundamental functions of this manual such as the account entry, inventory maintenance, blood requesting, and notification management shall be covered.*
  - ✓ *Testable: A non-technical user will go through the user manual to ensure that it is clear and complete.*
- VBB-TRN-002: *The system will also have a first run on boarding tutorial (e.g., a sequence of guided screens or brief animated tours) that will familiarize new users with the initial features and navigation.*

- ✓ *Verifiable: The onboarding tutorial will be tested in terms of success in introducing new users to the basic operations of the app in UAT.*
- *VBB-TRN-003: The system will contain context-sensitive help snippets or tooltips on the complicated input boxes or less user-friendly features.*
- ✓ *Verifiable: The existence of help features will be verified in all the complex elements identified.*

### 3.8.2 Packaging Requirements

- *VBB-PKG-001: The mobile app will be packaged to be distributed via the Android devices, via Google Play Store, and must be packaged according to Google packaging and submissions instructions (ex: AAB - Android App Bundle format).*
- *VBB-PKG-002: The mobile application is to be distributed as an enterprise or perhaps through the apple App Store on the iOS devices, according to the packaging and submission rules of Apple (e.g. IPA format).*
- *VBB-PKG-003 The backend server application will be packaged in the form of Docker containers, including all the required dependencies to allow easy deployment on cloud resources.*
- *Testable: Verifiable: Docker images will be developed and correctly deployed to a test environment.*
- *VBB-PKG-004: The source code repository will contain a top-level README.md file with the instructions on how to start the project, how to build it, and a general architectural overview of the project.*
- *VBB-PKG-005: The code repository will contain required configuration files (e.g., requirements.txt in case of Python, package.json in case of frontend, docker-compose.yml) to simplify the setting up of the environment and rebuilding.*

### 3.8.3 Legal Requirements

- *VBB-LEG-001: The system will have an End-User License Agreement (EULA), which the user is required to accept before the first use that will include the terms of service, data privacy practices, and limitations on use.*
- *Verifiable: EULA presented and acceptance by the user made during the first app setup.*
- *VBB-LEG-002: The system will clearly indicate all the available copyright data (e.g., project name, year, organization) in the section with the title About in the application.*
- *Verifiable: Copyright information displayed in the about section.*
- *VBB-LEG-003: The privacy policy of the data in the system will be easily visible in the application describing the methods of collecting, storing, processing, and protecting of user data and confidential information on health, according to Ethiopian laws on data protection.*
- *Verifiable: Privacy policy document is available through an in-app link.*
- *VBB-LEG-004: The system must contain open-source software which will be covered by a license that permits commercial or non-profit distribution (i.e. MIT, Apache 2.0, BSD), and that will be listed in the application or documentation.*
- *Checkable: The licenses of all the third party libraries will be checked and recorded.*

## 4. Change Management Process

*This section will outline the formal procedure to be used to deal with changes to this Software Requirements Specification (SRS) and subsequently to the scope and requirements of the project. Such*



*an organized change management process is essential to guarantee that every change is appropriately assessed, authorized, and communicated and is documented, thus keeping the definition of the system intact and consistent.*

## 4.1 Change Submission

- *VBB-CHG-001: A change request can be submitted by any stakeholder (e.g., Project Manager, Development Lead, QA Lead, a representative of a healthcare facility, EBTBS, or , etc.).*
- *VBB-CHG-002: The change requests will be presented formally with the help of a dedicated Change Request (CR) form or an equivalent feature of a project management tool (e.g., Jira, Trello, GitHub Issues).*
- *VBB-CHG-003: Every change request will present:*
  - ✓ *A CR ID (generated by the system or inserted by the person).*
  - ✓ *Date of submission.*
  - ✓ *Name of the submitter.*
  - ✓ *Overview of the intended change.*
  - ✓ *Rationale behind change (e.g. new regulatory requirement, better user experience, defect found, scope clarification).*
  - ✓ *Impact assessment (preliminary estimate of effort, expenditure, schedule, and requirements/components impacted).*
  - ✓ *Priority suggested (e.g. Critical, High, Medium, Low).*
  - ✓ *Identification of section(s) or requirement ID(s) in SRS that are affected.*
  - ✓ *As per the guideline, Change Evaluation and Approval take place at the fourth stage of the change management process. According to the guideline, Change Evaluation and Approval occur at the fourth level of change management process.*
- *VBB-CHG-004: All of the change requests should be examined by the Project Manager as complete and preliminary impacted at the time of submission.*
- *VBB-CHG-005: A Change Control Board (CCB) will be created, which will be made up of the Project Manager, Technical Lead, a key representative of the main customer/stakeholder group (e.g. Health Official) and a QA Lead.*
- *VBB-CHG-006: The CCB will meet frequently (e.g. once a week or once in three weeks) to discuss outstanding change requests.*
- *VBB-CHG-007: The CCB will consider every change request according to its effect on scope, schedule, budget, and technical feasibility and adherence to project goals.*
- *VBB-CHG-008: The CCB will accept, rebuff, postpone, or demand additional information on each change request.*
- *VBB-CHG-009: The CCB must approve a change request by a consensus decision, or a majority decision as an alternative to a consensus decision, if a consensus cannot be reached with the Project Manager having a tie-breaking power.*
- *VBB-CHG-010: All the decisions made by CCB (approved, rejected, deferred) should be mustered and the supporting reasons, date, and parties upon which the decision was made.*

## 4.2 Implementation of Change and Communication.

- *VBB-CHG-011: Approved change requests The Project Manager shall make update on the corresponding sections of this SRS document.*

- 
- VBB-CHG-012: Every change of an item in the SRS will contain:
    - ✓ The original requirement ID.
    - ✓ The date of modification.
    - ✓ The CR ID which led to change.
    - ✓ A short outline of the change.
    - ✓ The amended requirement text.
  - VBB-CHG-013: The SRS document will have a new version issued in case of a major change, and all the stakeholders should be informed of the availability of a new version.
  - VBB-CHG-014: The Project Manager will inform all the impacted members of the project team and project stakeholders about the impact of approved changes.
  - VBB-CHG-015: The QA team shall verify against updated SRS set against a change implemented in the software after implementing the change in the software.

## References

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## A. Appendices

This appendix adds on to the requirements that were described in the main body of this SRS. It contains the analysis of the target users and stakeholders, reasons why the core problem needs to be addressed, and conceptual models as a result of the first project planning stage.

Appendix A: Stakeholder and User Analysis.

This part describes the users and stakeholders of the Virtual Blood Bank system that give context to the functional and non-functional requirements.

### A.1. Primary User Persona (Indirect Beneficiary)

This character is the final beneficiary of the system whose life and health results are expected to be enhanced by the project. It is essential to know her context in order to appreciate the significance of the application.

**Demographics Gender:** Female Ages: 20-30 Rural Ethiopia (e.g., Tula) Education Illiterate or little formal education.

#### ❖ Significant Contention areas (Pain Points)

- ✓ Low levels of technology.

- ✓ Lack of access to health information.
- ✓ The cultural practices can be a deterrent to hospital care.
- ✓ Health expenses and travelling are a huge burden since family income is low.
- ✓ Primary Objective To deliver a safe child and provide a healthy life to her baby and the family.
- ✓ Motivations The solution should be believed and proved to save lives in her community.
- ✓ The solution must conform to cultural and religious standards.

## A.2 Key Stakeholders

The main stakeholders who are going to be directly affected by the project or my special interest towards the project are the following groups.

1. **Mothers:** Rural Ethiopian women in low-income households. Their living conditions, well-being and security are the key focus.
2. **Families:** The immediate and extended family members that are affected emotionally and economically due to the death of maternal mortality.
3. **Healthcare Institutions:** Rural clinics and hospitals with poor resources, staffing and logistical issues. It is a system that is aimed at enhancing their efficiency and safety.
4. **Government:** Ethiopian ministry of health and local government agencies in charge of healthcare policy, funding and positive outcomes.

## A.3 User Interview Insights

These are the main lessons that were learned when interviewing members of the community and institutional stakeholders. These were the combined views that led into the problem definition core and technical and functional requirements of this system.

### Community Perspective

Note: This refers to the insights that were realized when the community members were interviewed, such as

- Lemlem Lema,
- Housewife,
- Age 35.

The official maternal mortality rate that is caused by postpartum bleeding is probably underreported and the number of deaths is much greater.

Rural hospitals often simply cannot afford to offer even the simplest of services, and blood transfusion is a highly pressing, but usually unavailable, resource.

The current state of maternal healthcare services is highly and widely met with community frustration.

### Institutional and Technical Perspective.

- ❖ The information obtained during an interview with
  - Merdokiyos Mano,
  - The Chief Director of the IT Department at the Ethiopian Blood and Tissue Bank Service
  - Phone\_No:+251911580854

The national blood management system is centralized and as it undergoes the process of modernization, it is experiencing serious logistical problems in terms of delivering blood to the remote and rural systems in time.

One of the recent projects of the Ethiopian Blood and Tissue Bank Service is the active construction of the Blood Safety Information System (BSIS), which is one of the significant current initiatives aimed at enhancing blood safety, traceability, and macro-level inventory management on a national level.

All new digital health solution should follow certain rules and regulations set in the country regarding blood storage, handling, data privacy, and inter-facility transfer.

Although the BSIS tackles the national (top-down) infrastructure, it is understood that such solutions need to be complemented by localized (bottom-up) solutions such as the Virtual Blood Bank to resolve the last-mile (real-time) issue of communication and sharing of resources between individual hospitals on a district basis.

#### **Appendix B: Justification and Supporting Data on the problem.**

In this section, the main problem statement and information are presented that can substantiate the necessity of this system.

##### **B.1 Core Problem Statement**

Maternal death rate in rural Ethiopia is also very high with an official figure recording 401 deaths per 100,000 live births. One of the main causes of this crisis is postpartum haemorrhage (PPH) that predisposes more than a quarter of these deaths. This is a matter that is severely exasperated in remote areas where access to blood to carry out transfusion is low or nonexistent and this has resulted in avoidable fatalities and gross family hardships. The current project fits the gap directly as it provides a system that enhances access to the current blood supplies through better logistics in the case of obstetric emergencies.

##### **B.2 Supporting References**

The statistic of 401/100,000 of the Ethiopian Ministry of Health / WHO of 2016 or later official reports on the Ethiopian population. (Note: This should be fully and formally cited here).

